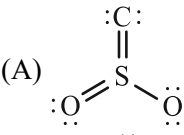
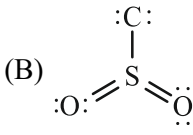
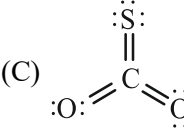


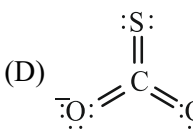
DPP-01_Chemical Bonding_Chemistry_Class 10th

- In which one of the following compounds does the central atom obey the octet rule?
(A) XeF_4 (B) XeOF_2
(C) SCl_2 (D) AlCl_3
- The attractive force which holds various constituents together in different chemical species is called
(A) valence bond (B) chemical bond
(C) atomic bond (D) electrovalent bond
- The octet rule is observed in
(A) PCl_5 (B) CO_2
(C) BCl_3 (D) SF_6
- Which one of the following contains ionic, covalent and co-ordinate bonds?
(A) NaOH (B) NaCl
(C) HCl (D) NaNCl
- Which of the following is correct for the process of formation of a chemical bond?
(A) Electron-electron repulsion becomes more than the nucleus-electron attraction.
(B) Energy of the system does not change.
(C) Energy increases.
(D) Energy decreases.
- Example of super octet molecule is
(A) SF_6 (B) PCl_5
(C) IF_7 (D) All of these
- The number of electrons involved in the bond formation in N_2 molecule is
(A) 2 (B) 4
(C) 10 (D) 6
- The octet rule is not obeyed in
(A) CO_2 (B) BCl_3
(C) PCl_5 (D) (B) and (C) both
- NH_3 and BF_3 combine readily because of the formation of
(A) a covalent bond (B) a hydrogen bond
(C) a coordinate bond (D) an ionic bond
- Which of the following species are hypervalent?
1. ClO_4^- 2. BF_3
3. SO_4^{2-} 4. CO_3^{2-}
(A) 1, 2, 3 (B) 1, 3
(C) 3, 4 (D) 1, 2
- Boron forms covalent compound due to
(A) Small size
(B) Higher ionisation energy
(C) Lower ionisation energy
(D) Both (A) and (B)
- The covalency of nitrogen in HNO_3 is
(A) 0 (B) 3
(C) 4 (D) 5
- Which of the following is observed in metallic bonds?
(A) Mobile valence electrons
(B) Localised electrons
(C) Highly directed bond
(D) None of these
- To which of the following species octet rule is applicable:
(A) BrF_5 (B) SF_6
(C) IF_7 (D) CO
- The possible structure of monothiocarbonate ion is:

(A) 

(B) 

(C) 

(D) 

DPP-01_Chemical Bonding_Chemistry_Class 10th

ANSWER KEY

1. (C)
2. (B)
3. (B)
4. (D)
5. (D)
6. (D)
7. (D)
8. (D)

9. (C)
10. (B)
11. (D)
12. (C)
13. (A)
14. (D)
15. (D)



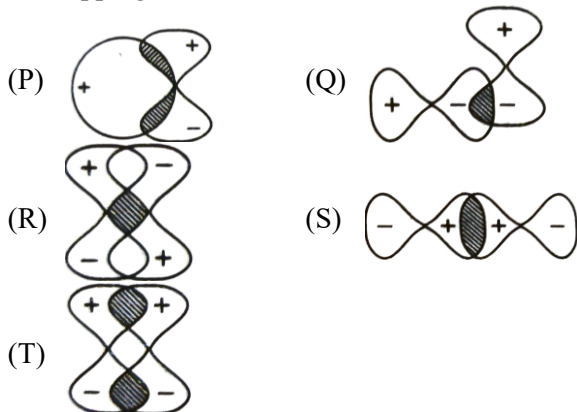
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DPP-02_Chemical Bonding_Chemistry_Class 10th

1. Which one of the following molecules are formed by p-p overlapping?

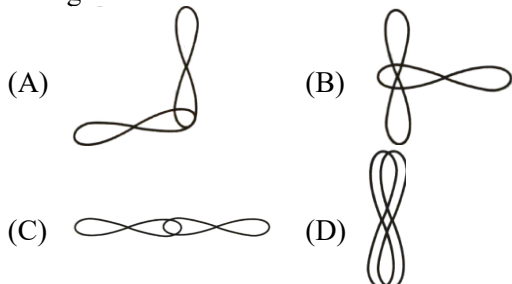
(A) Cl_2 (B) HCl
(C) H_2O (D) NH_3

2. Which of the following atomic orbital overlappings are not allowed?



(A) All of these (B) (P) (Q) (R)
(C) (P) (R) (T) (D) (Q) only

3. Which p-orbitals overlapping would give the strongest bond?



4. The bond in the formation of fluorine molecule will be:

(A) Due to s-s overlapping
(B) Due to s-p overlapping
(C) Due to p-p overlapping
(D) Due to hybridization

5. Which type of overlapping results the formation of a π -bond

(A) Axial overlapping of s-s orbitals
(B) Lateral overlapping of p-p orbitals
(C) Axial overlapping of p-p orbitals
(D) Axial overlapping of s-p orbitals

6. In a sigma bond

(A) Sidewise as well as end to end overlap of orbitals take place
(B) Sidewise overlap of orbitals takes place
(C) End to end overlap of orbitals takes place
(D) None of the above

7. Which is correct option for overlapping strength

(A) $2s-2s < 2s-2p < 2p-2p$
(B) $1s-1s < 1s-2p < 1p-2p$
(C) $2s-2s < 1s-2p < 1s-1s$
(D) Both (A) and (C)

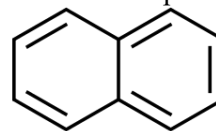
8. Which overlapping consist of maximum overlapped area between nucleus:

(A) $2s-2s$
(B) $2p-2p$
(C) $2s-2p$
(D) Both (A) and (C)

9. Among the following which species has same number of σ and π -bond?

(A) C_7H_8 (B) $\text{C}_2(\text{CN})_4$
(C) C_2H_4 (D) $\text{HC} \equiv \text{CH}$

10. Number of π -bonds in naphthalene is:

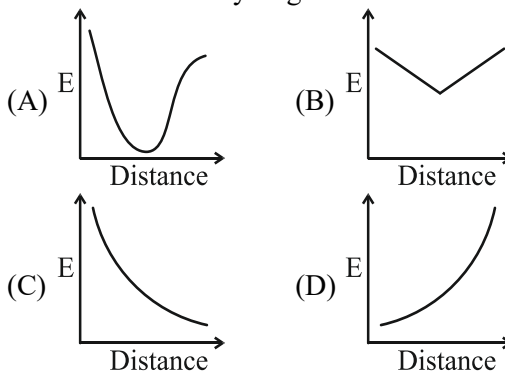


(A) 6 (B) 3
(C) 4 (D) 5

11. Ratio of the σ and π bonds is maximum in

(A) naphthalene
(B) tetracyano methane
(C) enolic form of urea
(D) equal

12. Which plot best represents the potential energy (E) of two hydrogen atoms as then approach one another to form a hydrogen molecule?



13. Main axis of a diatomic molecule is z, molecular orbitals p_x and p_y overlap to form, which of the following orbital?

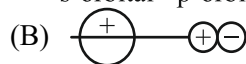
(A) π -molecular orbital
(B) σ -molecular orbital
(C) δ -molecular orbital
(D) no bond will form

14. Which of the following leads to bonding?

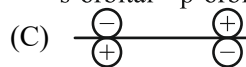
s-orbital p-orbital



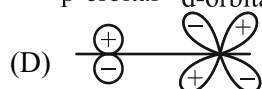
s-orbital p-orbital



s-orbital p-orbital



p-orbital d-orbital



15. Which cannot be explained by VBT?

- (A) Overlapping
- (B) Bond formation
- (C) Paramagnetic nature of oxygen
- (D) Shapes of molecules

DPP-02_Chemical Bonding_Chemistry_Class 10th

ANSWER KEY

1. (A)
2. (B)
3. (C)
4. (C)
5. (B)
6. (C)
7. (D)
8. (B)

9. (B)
10. (D)
11. (C)
12. (A)
13. (D)
14. (B)
15. (C)



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DPP-03_Chemical Bonding_Chemistry_Class 10th

- The percentage of s -character in the hybridized orbitals of B in BF_3 is:
(A) 25 (B) 50
(C) 75 (D) 33.3
- In which of the following pairs both the species have sp^3 hybridization?
(A) $\text{SiF}_4, \text{BeH}_2$ (B) $\text{NF}_3, \text{H}_2\text{O}$
(C) NF_3, BF_3 (D) $\text{H}_2\text{S}, \text{BF}_3$
- Which of the following statements is correct?
(A) sp^3 hybrid orbitals have equal s and p character.
(B) The bond angle decreases with the decrease of s character of hybridized orbital.
(C) Resonance decreases the stability of a molecule.
(D) Resonance is due to delocalization of sigma electrons.
- The percentage s -character of the hybrid orbitals in methane, ethene and ethyne are respectively:
(A) 25,33,50 (B) 25,50,75
(C) 50,75,100 (D) 10,20,40
- The percentage of p -character of the hybrid orbitals in graphite and diamond are respectively:
(A) 33 and 25 (B) 50 and 75
(C) 67 and 75 (D) 33 and 75
- Pick out the incorrect statement from the following:
(A) sp hybrid orbitals are equivalent and are at an angle of 180° with each other.
(B) sp^2 hybrid orbitals are equivalent and bond angle between any two of them is 120°
(C) sp^3d^2 hybrid orbitals are equivalent and are oriented towards corners of a regular octahedron.
(D) sp^3d^3 hybrid orbitals are not equivalent.
- The hybridization of carbon in molecule CO_2 is:
(A) sp (B) sp^2
(C) sp^3 (D) sp^3d
- What is the type of hybridisation found in methane?
(A) sp^3 (B) sp^2
(C) sp^1 (D) None of these
- The hybridization at carbons involved in C^*-C^* bond in $\text{HC}\equiv\text{C}^*-\text{CH}=\text{CH}_2$ is:
(A) sp^3-sp^2 (B) sp^3-sp^3
(C) $sp-sp^2$ (D) $sp-sp^3$
- The hybridisation and shape of ClO_3^- is:
(A) sp^3 and tetrahedral
(B) sp^3 and triangular pyramidal
(C) sp^3 and triangular planar
(D) sp^3d and trigonal bipyramidal
- The state of hybridization of Xe in XeF_4 is:
(A) sp^2 (B) sp^3
(C) sp^3d (D) sp^3d^2
- sp^2 -hybridization is shown by:
(A) BeCl_2 (B) BF_3
(C) NH_3 (D) XeF_2
- Carbon atoms in $\text{C}_2(\text{CN})_4$ are:
(A) sp -hybridized
(B) sp^2 -hybridized
(C) sp and sp^2 -hybridized
(D) sp, sp^2 and sp^3 -hybridized
- Which one of the following is the correct set with respect to molecule, hybridization and shape?
(A) BeCl_2, sp^2 , linear
(B) BeCl_2, sp^2 , triangular planar
(C) BCl_3, sp^2 , triangular planar
(D) BCl_3, sp^3 , tetrahedral
- The hybridization of central iodine atom in $\text{IF}_5, \text{I}_3^-$ and I_3^+ are respectively:
(A) sp^3d^2, sp^3d, sp^3 (B) sp^3d, sp^3d, sp^3
(C) sp^3d^2, sp^3d^2, sp^3 (D) sp^3d, sp^3d^2, sp^3

DPP-03_Chemical Bonding_Chemistry_Class 10th

ANSWER KEY

1. (D)
2. (B)
3. (B)
4. (A)
5. (C)
6. (D)
7. (A)
8. (A)

9. (C)
10. (B)
11. (D)
12. (B)
13. (C)
14. (C)
15. (A)



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DPP-04_Chemical Bonding_Chemistry_Class 10th

1. Which of the following molecules has more than one lone pair on central atom?
(A) SO_2 (B) XeF_2
(C) SiF_4 (D) CH_4
2. In the structure of ClF_3 , the number of lone pairs of electrons on central atom 'Cl' is:
(A) one (B) two
(C) four (D) three
3. The central atom of which one of the following molecules, has 3 bond pairs and 2 lone pairs of electrons in its valence shell?
(A) Boron trifluoride
(B) Phosphorous trichloride
(C) Chlorine trifluoride
(D) Xenon trioxide
4. Number of b.p. and l.p. in NO_3^- is:
(A) 3 b.p. + 1 l.p. (B) 4 b.p. + 0 l.p.
(C) 2 b.p. + 2 l.p. (D) 1 b.p. + 1 l.p.
5. SF_4 can be classified as
(A) AB_4 type (B) AB_4E type
(C) AB_4E_2 type (D) None of these
6. In which of the following molecules, the central atom has two lone pairs of electrons?
(A) SF_4 (B) BrF_5
(C) SO_2 (D) XeF_4
7. Among the following molecules: SO_2 , SF_4 , ClF_3 , BrF_5 and XeF_4 which of the following shapes does not describe any of the molecules mentioned?
(A) Bent
(B) Trigonal bipyramidal
(C) See-saw
(D) T-shape
8. In which of the following molecule/ion, all the bonds are not equal?
(A) XeF_4 (B) BF_4^-
(C) C_2H_4 (D) SiF_4
9. Incorrect matching amongst the following is:
(i) Linear - H_2O , SO_2
(ii) V-shaped - $:\text{CH}_2$, SnCl_2
(iii) See Saw - SF_4 , TeCl_4
(iv) T-shaped - ICl_3 , ClF_3
(A) only (i)
(B) both (ii and (iii)
(C) only (iii)
(D) both (iii) and (iv)
10. According to VSEPR theory molecule having T-shape is:
(A) ClF_3 (B) NH_3
(C) BF_3 (D) H_2O
11. Which of the following has a regular geometry?
(A) CHCl_3 (B) PCl_3
(C) XeF_6 (D) SF_4
12. **Assertion:** ClF_3 has T-shape structure.
Reason: It has two lone pairs arranged at 180° angle.
(A) If both assertion and reason are true and reason is the correct explanation of assertion.
(B) If both assertion and reason are true but reason is not the correct explanation of assertion.
(C) If assertion is true but reason is false.
(D) If both assertion and reason are false.
13. Which of the following molecules is linear in shape?
(A) XeF_6 (B) XeO_3
(C) XeF_4 (D) XeF_2
14. The shape of BeCl_2 is:
(A) Linear (B) Bent
(C) T-shaped (D) None of these
15. Which of the following is planar?
(A) XeO_4 (B) XeO_2F_2
(C) XeO_3 (D) XeF_4

DPP-04_Chemical Bonding_Chemistry_Class 10th

ANSWER KEY

1. (B)
2. (B)
3. (C)
4. (B)
5. (B)
6. (D)
7. (B)
8. (C)

9. (A)
10. (A)
11. (A)
12. (C)
13. (D)
14. (A)
15. (D)



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DPP-05_Chemical Bonding_Chemistry_Class 10th

1. The species, having bond angles of 120° is
 (A) ClF_3 (B) NCl_3
 (C) BCl_3 (D) PH_3
2. Consider the molecules CH_4 , NH_3 and H_2O . Which of the given statements is false?
 (A) The $\text{H}-\text{O}-\text{H}$ bond angle in H_2O is smaller than the $\text{H}-\text{N}-\text{H}$ bond angle in NH_3 .
 (B) The $\text{H}-\text{C}-\text{H}$ bond angle in CH_4 is larger than the $\text{H}-\text{N}-\text{H}$ bond angle in NH_3 .
 (C) The $\text{H}-\text{C}-\text{H}$ bond angle in CH_4 , the $\text{H}-\text{N}-\text{H}$ bond angle in NH_3 , and the $\text{H}-\text{O}-\text{H}$ bond angle in H_2O are all greater than 90° .
 (D) The $\text{H}-\text{O}-\text{H}$ bond angle in H_2O is larger than the $\text{H}-\text{C}-\text{H}$ bond angle in CH_4 .
3. The molecule having smallest bond angle is:
 (A) AsCl_3
 (B) SbCl_3
 (C) PCl_3
 (D) NCl_3
4. Which has the least bond angle:
 (A) NH_3 (B) BeF_2
 (C) H_2O (D) CH_4
5. The correct order of bond angles is:
 (A) $\text{NO}_2^- > \text{NO}_2^+ > \text{NO}_2$
 (B) $\text{NO}_2^+ > \text{NO}_2^- > \text{NO}_2$
 (C) $\text{NO}_2 > \text{NO}_2^+ > \text{NO}_2^-$
 (D) $\text{NO}_2^+ > \text{NO}_2 > \text{NO}_2^-$
6. The correct order of $\text{H}-\text{M}-\text{H}$ bonds angle is:
 (A) $\text{NH}_3 < \text{PH}_3 < \text{SbH}_3 < \text{BiH}_3$
 (B) $\text{AsH}_3 < \text{SbH}_3 < \text{PH}_3 < \text{NH}_3$
 (C) $\text{NH}_3 < \text{PH}_3 < \text{BiH}_3 < \text{SbH}_3$
 (D) $\text{BiH}_3 < \text{SbH}_3 < \text{AsH}_3 < \text{PH}_3$
7. The correct order of bond angles (smallest first) in H_2S , NH_3 , BF_3 and SiH_4 is
 (A) $\text{H}_2\text{S} < \text{SiH}_4 < \text{NH}_3 < \text{BF}_3$
 (B) $\text{H}_2\text{S} < \text{NH}_3 < \text{BF}_3 < \text{SiH}_4$
 (C) $\text{H}_2\text{S} < \text{NH}_3 < \text{SiH}_4 < \text{BF}_3$
 (D) $\text{NH}_3 < \text{H}_2\text{S} < \text{SiH}_4 < \text{BF}_3$
8. In a covalent bond formation,
 (A) transfer of electrons takes place
 (B) equal sharing of electrons between two atoms takes place
 (C) electrons are shared by one atom only
 (D) electrons are donated by one atom and shared by both atoms.
9. In which of the following species the bond is non-directional?
 (A) NCl_3 (B) RbCl
 (C) BeCl_2 (D) BCl_3
10. Sodium chloride has a crystalline structure made up of Na^+ and Cl^- ions. Why does NaCl not conduct electricity in solid state?
 (A) The ions of NaCl become mobile only in molten state and are not free to move in solid state
 (B) The crystalline structure does not have ions
 (C) When a bond is formed between ions they lose their charges
 (D) None of these
11. Most ionic compounds have:
 (a) High melting point and low boiling point
 (b) High melting point and non-directional bonds
 (c) High solubilities in polar solvents and low solubilities in non-polar solvents
 (d) Three dimensional network structures and are good conductors of electricity in the molten state
 (A) None of the above (B) b, c, d
 (C) a, b, c (D) All of the above
12. Which one of the following has the highest lattice energy?
 (A) LiCl (B) CaCl_2
 (C) LiF (D) KCl
13. Lattice energy of a solid increases if:
 (A) The radii of ions are small
 (B) The ions are neutral
 (C) Charges on the ions are small
 (D) Madelung constant decreases

14. Which is the Most soluble sulphate in water among the following compounds:

- (A) BaSO_4 (B) SrSO_4
(C) MgSO_4 (D) BeSO_4

15. The ionic compounds will be sparingly soluble if:

- (A) Hydration energy = lattice energy
(B) Hydration energy > lattice energy
(C) Hydration energy < lattice energy
(D) Hydration energy = 0

DPP-05_Chemical Bonding_Chemistry_Class 10th

ANSWER KEY

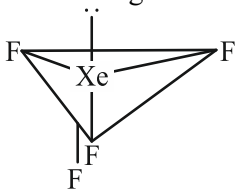
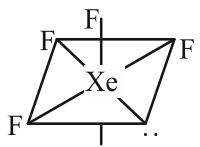
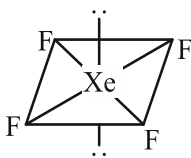
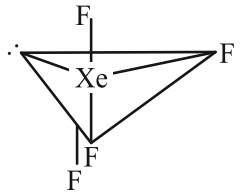
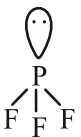
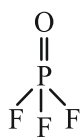
1. (C)
2. (D)
3. (B)
4. (C)
5. (D)
6. (D)
7. (C)
8. (B)

9. (B)
10. (A)
11. (B)
12. (B)
13. (A)
14. (D)
15. (A)



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DPP-06_Chemical Bonding_Chemistry_Class 10th

- Position of fluorine in PCl_3F_2
 (A) Both at equatorial position
 (B) Both at axial position
 (C) 1 at equatorial and 1 at axial position
 (D) Cannot be predicted
- In sp^3d hybridization with 2 lone pairs, the lone pairs are placed at _____ position and shape of molecule is _____.
 (A) Equatorial, trigonal bipyramidal
 (B) Axial, trigonal bipyramidal
 (C) Equatorial, Bent-T
 (D) Axial, Trigonal planar
- %s character in equatorial orbital in TBP is:
 (A) 50% (B) 25%
 (C) 33.33% (D) 0%
- Bond angle in PH_3 is:
 (A) Close to 109.28° (B) Same as NH_3
 (C) Close to 90° (D) Close to 120°
- Hybridisation of PBr_5 in the solid state is:
 (A) sp^3 (B) sp^3d
 (C) sp^2 (D) sp
- Each carbon in carbon suboxide is
 (A) sp^2 -hybridized
 (B) sp^3 -hybridized
 (C) sp -hybridized
 (D) sp^3 -hybridized but linked with one co-ordinate bond
- The type of overlapping which produces bond of maximum bond energy is:
 (A) sp^3-1s
 (B) sp^2-1s
 (C) $\text{sp}-1\text{s}$
 (D) All have same bond energy
- Which is the right structure of XeF_4 ?
 (A) 
 (B) 
 (C) 
 (D) 
- In which of the following central atom is unhybridized?
 (A) $\text{S}(\text{CH}_3)_2$ (B) SO_2
 (C) SiH_4 (D) PH_3
- Which of the following has maximum (C-C) length?
 (A) C_2H_2 (B) C_2H_4
 (C) C_2H_6 (D) $\text{C}_2\text{H}_2\text{Br}_2$
- In which of the following species, d-orbitals having xz and yz two nodal planes are involved in hybridisation of central atom?
 (A) IO_2F_2^- (B) ClF_4^-
 (C) IF_7 (D) None of these
- The molecule having bond pair in which % of s & p character is different from rest bond pairs.
 (A) SiF_4 (B) BCl_3
 (C) CF_3H (D) CCl_4
- Most basic among following is
 (A) NH_3 (B) PH_3
 (C) SbH_3 (D) BiH_3
- If θ_1 , θ_2 , θ_3 represents the $\angle\text{FPF}$ in I, II, III respectively, then
 I.  II.  III. 
 (A) $\theta_1 < \theta_2 < \theta_3$ (B) $\theta_1 < \theta_3 < \theta_2$
 (C) $\theta_3 = \theta_2 < \theta_1$ (D) $\theta_1 < \theta_2 = \theta_3$
- The longest C-C bond is present in which of the following compounds?
 (A) $\text{CH}_3 - \text{CF}_3$
 (B) $\text{FCH}_2 - \text{CH}_2\text{F}$
 (C) $\text{F}_2\text{CH} - \text{CHF}_2$
 (D) $\text{CF}_3 - \text{CF}_3$

DPP-06_Chemical Bonding_Chemistry_Class 10th

ANSWER KEY

1. (B)
2. (C)
3. (C)
4. (C)
5. (A)
6. (C)
7. (C)
8. (C)

9. (D)
10. (C)
11. (C)
12. (C)
13. (A)
14. (B)
15. (B)

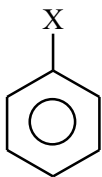


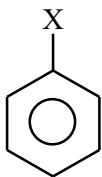
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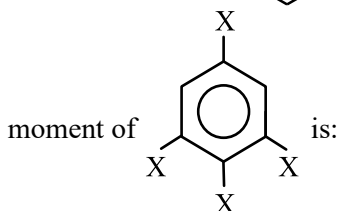
DPP-07_Chemical Bonding_Chemistry_Class 10th

1. Zero dipole moment is possessed by:

- (A) PCl_3 (B) BF_3
(C) ClF_3 (D) NH_3



2. Dipole moment of  is 1.5D. The dipole



- (A) 1.5 D (B) 2.25 D
(C) 1 D (D) 3 D

3. Which one of the following is a non-polar molecule?

- (A) CCl_4 (B) CHCl_3
(C) CH_2Cl_2 (D) CH_3Cl

4. Identify the non-polar molecule in the set of compounds given:

HCl , HF , H_2 , HBr

- (A) H_2 (B) HCl
(C) HF , HBr (D) HBr

5. The bond angle and dipole moment of water respectively, are:

- (A) 109.5° , 1.84 D (B) 107.5° , 1.56 D
(C) 104.5° , 1.84 D (D) 102.5° , 1.56 D

6. Which of the following has dipole moment?

- (A) CO_2 (B) p-dichlorobenzene
(C) NH_3 (D) CH_4

7. The decreasing order of dipole moment of molecules is

- (A) $\text{NF}_3 > \text{NH}_3 > \text{H}_2\text{O}$ (B) $\text{NH}_3 > \text{NF}_3 > \text{H}_2\text{O}$
(C) $\text{H}_2\text{O} > \text{NH}_3 > \text{NF}_3$ (D) $\text{H}_2\text{O} > \text{NF}_3 > \text{NH}_3$

8. H_2O has a net dipole moment while BeF_2 has zero dipole moment because:

- (A) F is more electronegativity than oxygen
(B) Be is more electronegativity than oxygen
(C) H_2O molecule is linear and BeF_2 is bent
(D) BeF_2 molecule is linear and H_2O is bent

9. BeCl_2 has zero dipole moment whereas H_2O has dipole moment because:

- (A) Water is linear
(B) H_2O is bent
(C) F is more electronegative than O
(D) Hydrogen bonding is present in H_2O

10. Which of the following species are polar:

- (1) C_6H_6 (2) XeF_2
(3) SO_2 (4) SF_4
(5) SF_6

Correct answer is:

- (A) (2) & (4)
(B) (1), (2) & (5)
(C) (1) & (5)
(D) (3) & (4)

11. Dipole moment is shown by:

- (A) 1, 4-Dichlorobenzene
(B) Cis 1, 2-dichloroethene
(C) Trans -1, 2-dichloro ethene
(D) Benzene

12. What conclusion can be draw from the fact that BF_3 has no dipole moment but PF_3 does:

- (A) BF_3 is not symmetrical but PF_3 is symmetrical
(B) BF_3 molecule must be linear
(C) Atomic radius of P is larger than that of B
(D) BF_3 molecule must be planar triangular

13. In which molecule are all atoms coplanar

- (A) CH_4 (B) BF_3
(C) PF_3 (D) NH_3

14. Which of the following is the correct order of dipole moment?

- (A) $\text{NH}_3 < \text{BF}_3 < \text{NF}_3 < \text{H}_2\text{O}$
(B) $\text{BF}_3 < \text{NF}_3 < \text{NH}_3 < \text{H}_2\text{O}$
(C) $\text{BF}_3 < \text{NH}_3 < \text{NF}_3 < \text{H}_2\text{O}$
(D) $\text{H}_2\text{O} < \text{NF}_3 < \text{NH}_3 < \text{BF}_3$

15. Among the following molecules, which has the zero-dipole moment?

- (A) CH_4
(B) H_2O
(C) NF_3
(D) ClO_2

DPP-07_Chemical Bonding_Chemistry_Class 10th

ANSWER KEY

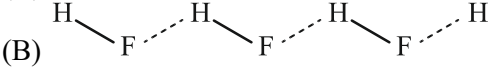
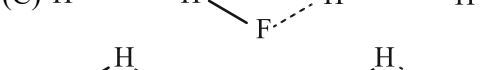
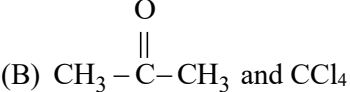
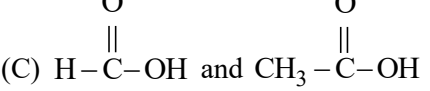
1. (B)
2. (A)
3. (A)
4. (A)
5. (C)
6. (C)
7. (C)
8. (D)

9. (B)
10. (D)
11. (B)
12. (D)
13. (B)
14. (B)
15. (A)



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DPP-08_Chemical Bonding_Chemistry_Class 10th

- For BF_3 molecule which of the following is true?
 - B-atom is sp^2 hybridised
 - There is a $\text{p}\pi\text{-p}\pi$ back bonding in this molecule
 - Observed B-F bond length is found to be less than the expected bond length
 - All of these
- Respective order of strength of back-bonding in boron trihalides is
 - $\text{BF}_3 < \text{BCl}_3 < \text{BBR}_3$
 - $\text{BF}_3 > \text{BCl}_3 > \text{BBR}_3$
 - $\text{BBR}_3 > \text{BCl}_3 > \text{BF}_3$
 - $\text{BF}_3 < \text{BCl}_3 < \text{BBR}_3$
- The nature of p-bonds in perchlorate ion is:
 - O ($\text{d}\pi$) - Cl ($\text{p}\pi$)
 - O ($\text{p}\pi$) - Cl ($\text{d}\pi$)
 - O ($\text{d}\pi$) - Cl ($\text{d}\pi$)
 - O ($\text{p}\pi$) - Cl ($\text{p}\pi$)
- The structure of diborane (B_2H_6) contains:
 - four ($2\text{c} - 2\text{e}^-$) bonds and two ($2\text{c} - 3\text{e}^-$) bonds
 - two ($2\text{c} - 2\text{e}^-$) bonds and two ($3\text{c} - 2\text{e}^-$) bonds
 - four ($2\text{c} - 2\text{e}^-$) bonds and four ($3\text{c} - 2\text{e}^-$) bonds
 - none
- The H bond in solid HF can be best represented as:
 - $\text{H}-\text{F} \cdots \text{H}-\text{F} \cdots \text{H}-\text{F}$
 - 
 - 
 - 
- The pairs of bases in DNA are held together by
 - Hydrogen bonds
 - Ionic bonds
 - Phosphate groups
 - Deoxyribose groups
- Gaseous SO_3 molecule
 - is planar triangular in shape with three σ -bonds from $\text{sp}^2\text{-p}$ overlap and three π -bonds formed by one $\text{p}\pi\text{-p}\pi$ overlap and two $\text{p}\pi\text{-d}\pi$ overlap.
 - is a pyramidal molecule with one double bond and two single bonds
 - planar triangular in shape with two double bonds between S and O and one single bond
 - is a planar triangular in shape with three σ bonds from $\text{sp}^2\text{-p}$ overlap and three π -bonds formed by two $\text{p}\pi\text{-p}\pi$ overlap and one $\text{p}\pi\text{-d}\pi$ overlap.
- Intramolecular hydrogen bonding is found in:
 - Salicylaldehyde
 - Water
 - Acetaldehyde
 - Phenol
- Which of the following compounds does not contain -1 formal charge on any of O atom:
 - O_3
 - H_3PO_4
 - HNO_3
 - N_2O_5
- NH_3 and BF_3 combine readily because of the formation of:
 - a covalent bond
 - a hydrogen bond
 - a coordinate bond
 - an ionic bond
- Intermolecular hydrogen bond is present in which of the following pair of molecules?
 - SiH_4 and SiF_4
 - 
 - 
 - CH_3OCH_3 and H_2O_2
- In which of the following compounds B-F bond length is shortest?
 - BF_4^-
 - $\text{BF}_3 \rightarrow \text{NH}_3$
 - BF_3
 - $\text{BF}_3 \leftarrow \text{N}(\text{CH}_3)_3$
- The order of strength of hydrogen bonds is:
 - $\text{ClH} \cdots \text{Cl} > \text{NH} \cdots \text{N} > \text{OH} \cdots \text{O} > \text{FH} \cdots \text{F}$
 - $\text{ClH} \cdots \text{Cl} < \text{NH} \cdots \text{N} < \text{OH} \cdots \text{O} < \text{FH} \cdots \text{F}$
 - $\text{ClH} \cdots \text{Cl} < \text{NH} \cdots \text{N} > \text{OH} \cdots \text{O} > \text{FH} \cdots \text{F}$
 - $\text{ClH} \cdots \text{Cl} < \text{NH} \cdots \text{N} < \text{OH} \cdots \text{O} > \text{FH} \cdots \text{F}$
- The correct order of boiling point is:
 - $\text{H}_2\text{O} < \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$
 - $\text{H}_2\text{O} > \text{H}_2\text{Se} < \text{H}_2\text{Te} < \text{H}_2\text{S}$
 - $\text{H}_2\text{O} > \text{H}_2\text{S} > \text{H}_2\text{Se} > \text{H}_2\text{Te}$
 - $\text{H}_2\text{O} > \text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S}$
- Which of the following statements is false for trisilylamine?
 - Three sp^2 orbitals are used for a σ bonding, giving a plane triangular structure.
 - The lone pair of electrons occupy as p-orbital at right angles to the plane triangle and this overlaps with empty p-orbitals on each of the three silicon atoms resulting in π bonding.
 - The N-Si bond length is shorter than the expected N-Si bond length.
 - It is a weaker Lewis base than trimethyl amine.

DPP-08_Chemical Bonding_Chemistry_Class 10th

ANSWER KEY

1. (D)
2. (B)
3. (B)
4. (D)
5. (C)
6. (A)
7. (A)
8. (A)

9. (B)
10. (C)
11. (C)
12. (C)
13. (B)
14. (D)
15. (B)



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DPP-09_Chemical Bonding_Chemistry_Class 10th

- The bond length in O_2^+ , O_2 , O_2^- and O_2^{2-} follows the order:
 (A) $O_2^{2-} > O_2^- > O_2 > O_2^+$
 (B) $O_2^+ > O_2 > O_2^- > O_2^{2-}$
 (C) $O_2 > O_2^- > O_2^{2-} > O_2^+$
 (D) $O_2^- > O_2^{2-} > O_2^+ > O_2$
- Which of the following of pair species has the bond order equal to 1/2?
 (A) H_2 , H_2^+ (B) He_2^+ , He_2
 (C) He_2 , H_2^+ (D) H_2^+ , He_2^+
- The correct statement among the following is
 (A) N_2 has higher bond order than N_2^+ and hence has larger bond length compared to N_2^+
 (B) N_2^+ has higher bond order than N_2 and hence has larger bond length compared to N_2
 (C) N_2 has higher bond order than N_2^+ and hence has higher dissociation energy compared to N_2^+
 (D) N_2 has lower bond order than N_2^+ and hence has lower dissociation energy compared to N_2^+ energy
- The highest occupied MO in N_2 and O_2^+ respectively are (take x-axis as internuclear axis)
 (A) $\sigma 2p_x$, $\pi^* 2p_y$ (B) $\pi 2p_y$, $\pi 2p_z$
 (C) $\sigma^* 2p_x$, $\sigma 2p_x$ (D) $\pi^* 2p_y$, $\pi^* 2p_z$
- A molecular orbital of a diatomic molecule changes sign when it is rotated by 180° around the molecular axis. This orbital is
 (A) σ (B) π
 (C) δ (D) ϕ
- Match the following items of column I with the appropriate items in column II.

Column I	Column II
Molecular species	Bond order and magnetic property
P. O_2^-	I. 2.5 paramagnetic
Q. O_2	II. 2.0 diamagnetic
R. O_2^+	III. 1.5 diamagnetic
S. O_2^{2-}	IV. 1.0 diamagnetic
	V. 2.0 paramagnetic
	VI. 1.5 paramagnetic

 (A) P-III, Q-V, R-IV, S-III
 (B) P-III, Q-II, R-I, S-IV
 (C) P-III, Q-V, R-IV, S-VI
 (D) P-VI, Q-V, R-I, S-IV
- The bond order of C_2 molecule is
 (A) 0 (B) 1
 (C) 2 (D) 3
- Which of the following has same bond order as NO^+ has?
 (A) CN^- (B) O_2^-
 (C) CN^+ (D) None of them
- Which of the following statements are correct?
 I. Bond order of NO is 2.5
 II. Bond order of NO^+ is 3.0
 III. Bond order of O_2 is 1.5
 IV. Bond order of CO is 3.0
 (A) I, II, III
 (B) I, II, IV
 (C) I, III, IV
 (D) II, III, IV
- According to MO theory which of the following statement about the magnetic character and bond order is correct regarding O_2^+ ?
 (A) Paramagnetic and bond order $< O_2$.
 (B) Paramagnetic and bond order $> O_2$.
 (C) Diamagnetic and bond order $< O_2$.
 (D) Diamagnetic and bond order $> O_2$.
- The correct order in which the O – O bond length increases in the following is:
 (A) $O_2 < O_3 < H_2O_2$
 (B) $H_2O_2 < O_3 < O_2$
 (C) $O_3 < O_2 < H_2O_2$
 (D) $O_2 < H_2O_2 < O_3$
- Which one of the following species is paramagnetic?
 (A) O_2^- (B) CN^-
 (C) CO (D) NO^+
- Molecular orbital electronic configuration for X_2^{n-} anion is:

$$KK^*(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p_x)^2(\pi 2p_y)^2$$

$$(\sigma 2p_z)^2(\pi^* 2p_x)^1$$
 The X_2^{n-} is:
 (A) N_2^- (B) O_2^-
 (C) N_2^{2-} (D) O_2^{2-}



14. Which of the following MO has two nodal planes perpendicular to each other

- (A) σ_{2s}
- (B) $\pi 2p_x$
- (C) $\pi^* 2p_y$
- (D) $\sigma^* 2p_z$

15. Which pair among the following has identical bond order?

- (A) CN^- , CO^+
- (B) O_2^- , O_2^+
- (C) CN^- , CN^+
- (D) NO^+ , CN^-

DPP-09_Chemical Bonding_Chemistry_Class 10th

ANSWER KEY

1. (A)
2. (D)
3. (C)
4. (A)
5. (B)
6. (D)
7. (C)
8. (A)

9. (B)
10. (B)
11. (A)
12. (A)
13. (A)
14. (C)
15. (D)



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DPP-10_Chemical Bonding_Chemistry_Class 10th

1. According to Fajans rule covalent bond is favoured by
(A) Small cation and large anion
(B) Small cation and small anion
(C) Large cation and large anion
(D) Large cation and small anion
2. Polarisation is the distortion of the shape of anion by an adjacently placed cation. Which of the following statement is correct?
(A) Maximum polarization is brought about by cation of high charge
(B) Maximum polarization is brought about by cation of Low charge
(C) A large cation is likely to bring about a large degree of polarization
(D) A small anion is likely undergo a large degree of polarization
3. Polarization power of a cation increases when
(A) Charge on the cation increases
(B) Size of the cation increases
(C) Charge on the cation decreases
(D) Has no relation to its size or charge
4. In an ionic compound $A^+ X^-$ the degree of covalent bonding is greatest when:
(A) A^+ and X^- ion are small
(B) A^+ and X^- ion are large
(C) A^+ and X^- ion are approximately of the same size
(D) A^+ is small and X^- is large
5. Amongst LiCl, RbCl, $BeCl_2$ and $MgCl_2$, the compounds with the greatest and the least ionic character respectively are:
(A) LiCl and RbCl (B) RbCl and $BeCl_2$
(C) RbCl and $MgCl_2$ (D) $MgCl_2$ and $BeCl_2$
6. The decreasing order of ionic nature of the following compound is:
(A) $LiI > NaBr > KCl > CsF$
(B) $LiI > KCl > NaBr > CsF$
(C) $CsF > NaBr > KCl > LiI$
(D) $CsF > KCl > NaBr > LiI$
7. The most covalent halide of s-Block is:
(A) CsF (B) CsBr
(C) BeI_2 (D) BeF_2
8. The correct order of polarizability of X^- is:
(A) $F^- > Cl^- > Br^- > I^-$ (B) $Cl^- > F^- > Br^- > I^-$
(C) $Br^- > Cl^- > F^- > I^-$ (D) $I^- > Br^- > Cl^- > F^-$
9. Which ion has a higher polarizing power?
(A) Mg^{2+} (B) Al^{3+}
(C) Ca^{2+} (D) Na^+
10. Pure phosphoric acid is very viscous, because.
(A) It is a strong acid
(B) It is tribasic acid
(C) It is hygroscopic
(D) It has PO_4^{3-} groups which are bonded by many hydrogens bonds
11. Least melting point is shown by the compound:
(A) $PbCl_2$ (B) $SnCl_4$
(C) NaCl (D) $AlCl_3$
12. Which of the following is arranged in order of increasing ionic character:
(A) $PbCl_2 < SnCl_4 < KCl < MgCl_2$
(B) $SnCl_4 < PbCl_2 < KCl < MgCl_2$
(C) $SnCl_4 < PbCl_2 < MgCl_2 < KCl$
(D) $PbCl_2 < SnCl_4 < MgCl_2 < KCl$
13. The ease of hydrolysis of trichlorides of group 15 elements in the order:
(A) $NCl_3 > PCl_3 > AsCl_3 > SbCl_3 > BiCl_3$
(B) $PCl_3 > NCl_3 > AsCl_3 > SbCl_3 > BiCl_3$
(C) $AsCl_3 > NCl_3 > PCl_3 > SbCl_3 > BiCl_3$
(D) $SbCl_3 > BiCl_3 > PCl_3 > NCl_3 > AsCl_3$
14. Which is most ionic:
(A) P_2O_5 (B) MnO
(C) CrO_3 (D) Mn_2O_7
15. Arrange the following in the increasing order of deviation from normal tetrahedral angle:
(A) $P_4 < PH_3 < H_2O$
(B) $PH_3 < H_2O < P_4$
(C) $P_4 < H_2O < PH_3$
(D) $H_2O < PH_3 < P_4$

DPP-10_Chemical Bonding_Chemistry_Class 10th

ANSWER KEY

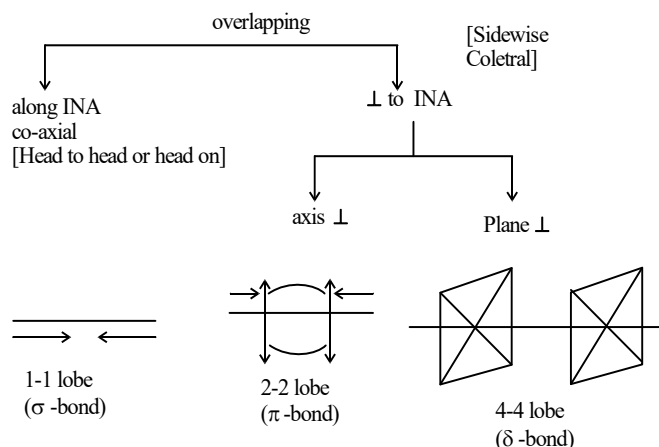
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2. (A)
3. (A)
4. (D)
5. (B)
6. (D)
7. (C)
8. (D)

9. (B)
10. (D)
11. (B)
12. (C)
13. (A)
14. (B)
15. (D)



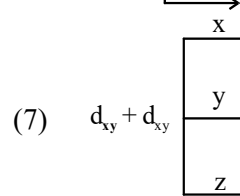
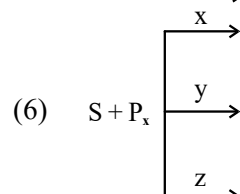
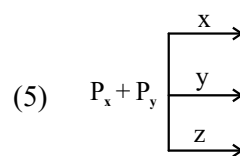
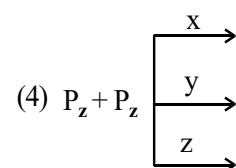
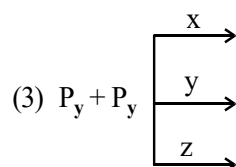
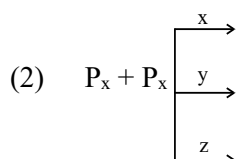
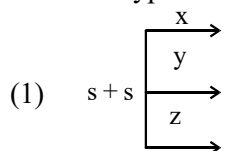
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DPP-11_Chemical Bonding_VBT(overlapping)_10th Class



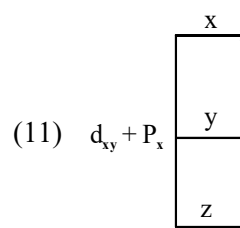
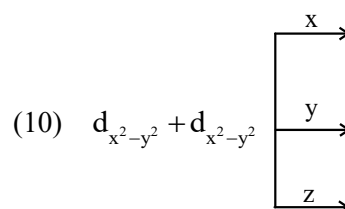
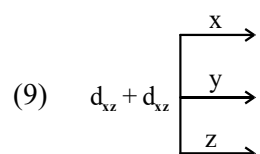
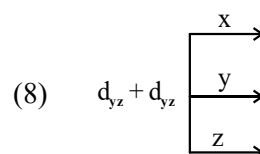
orbitals	no. of lobes	bond
s	0	σ
p	2	π
d	4	δ

Different types of Combinations:-



d_{xy} , d_{yz} , d_{zx} Form only π & δ

If INA Present in name of orbitals it will form δ -bond







DPP-12_Chemical Bonding_Chemistry(IOC)_10th Class

Molecule		Hydrolysed Product
II A Group		
1.	BeCl ₂	[Be(OH) ₂] _n + HCl
13th Group		
2.	BCl ₃	H ₃ BO ₃ + 3HCl
3.	BF ₃	H ₃ BO ₃ + 2H[BF ₄] (Partial hydrolysis due to complex formation)
14th Group		
4.	SiCl ₄	Si(OH) ₄ + 4HCl
5.	SiF ₄	Si(OH) ₄ + H ₂ [SiF ₆] (Partial hydrolysis due to complex formation)
6.	CCl ₄ (Superheated) [Via SN ¹]	COCl ₂ + 2HCl
15th Group		
7.	NCl ₃	3HOCl + 3NH ₃
8.	PCl ₃	H ₃ PO ₃ + 3HCl
9.	AsCl ₃	H ₃ AsO ₃ + 3HCl
10.	SbCl ₃ (Nither SN ¹ nor SN ²)	$\left. \begin{array}{l} \text{SbOCl} + 2\text{HCl} \\ \text{BiOCl} + 2\text{HCl} \end{array} \right\} \text{partial hydrolysi}$
11.	BiCl ₃ (Nither SN ¹ nor SN ²)	
12.	NF ₃ (Superheated) [Via SN ¹]	N ₂ O ₃
13.	PCl ₅ (Mixed SN ² + AE)	H ₃ PO ₄ + 5HCl
14.	POCl ₃ (AE)	H ₃ PO ₄
Oxides of nitrogen		
15.	N ₂ O ₃	HNO ₂
16.	NO ₂	HNO ₂ + HNO ₃
17.	N ₂ O ₅	HNO ₃
Oxides of phosphorous		
18.	P ₄ O ₁₀	4H ₃ PO ₄
19.	P ₄ O ₆	4H ₃ PO ₃
20.	P ₄ O ₈	H ₃ PO ₄ + H ₃ PO ₃
21.	P ₄ S ₁₀	4H ₃ PO ₄ + 10H ₂ S
16th Group		
22.	TeF ₆	H ₆ TeO ₆ + 6HF
23.	SeF ₆	H ₂ SeO ₄ + 6HF
24.	SO ₂ Cl ₂ (Add. Elimination mechanism)	H ₂ SO ₄ + 2HCl
25.	SOCl ₂ (Add. Elimination mechanism)	H ₂ SO ₃ + 2HCl
26.	SF ₆ SF ₆ (Via SN ¹)	does not hydrolyse at room temperature H ₂ SO ₄
27.	SF ₄	H ₂ SO ₃ + HF
28.	SOF ₄	H ₂ SO ₄ + 4HF

Oxides of sulphur		
29.	SO ₂ (Add. mechanism)	H ₂ SO ₃
30.	SO ₃ (Add. mechanism)	H ₂ SO ₄
Hydrolysis of oxyacids		
31.	H ₂ S ₂ O ₈	2H ₂ SO ₄ + H ₂ O ₂
32.	H ₂ SO ₅	H ₂ SO ₄ + H ₂ O ₂
33.	H ₄ P ₂ O ₇	2H ₃ PO ₄
34.	H ₄ P ₂ O ₅	2H ₃ PO ₃
35.	H ₄ P ₂ O ₆	H ₃ PO ₄ + H ₃ PO ₃
17th group		
Oxides of Chlorine		
36.	Cl ₂ O ₆	HClO ₃ + HClO ₄
37.	Cl ₂ O	2HOCl ₄
38.	Cl ₂ O ₇	2HClO ₄
Hydrolysis of interhalogen compounds		
39.	XX'	HX' + HOX
40.	XX ₃ ⁺	HXO ₂ + 3HX'
41.	XX ₅ '	HXO ₃ + 5HX'
42.	XX ₇ '	HXO ₄ + 7HX'
43.	F ₂	HF + O ₂
44.	Cl ₂	HOCl + HCl
45.	Br ₂	HOBr + HBr
Hydrolysis of xenon fluorides		
46.	XeF ₂ (Redox)	$\text{Xe} + \frac{1}{2}\text{O}_2 + 2\text{HF}$
47.	2XeF ₄ (Redox)	$\text{Xe} + \text{XeO}_3 + 8\text{HF} + \frac{1}{2}\text{O}_2$
48.	XeF ₆	XeO ₃ + 6HF
49.	$\begin{array}{ccc} \text{XeF}_6 & \xrightarrow{\text{(in basic medium)}} & \text{HXeO}_4^- \\ \downarrow \text{H}_2\text{O} & \nearrow \text{OH}^- & \uparrow \text{disproportionation} \\ \text{XeO}_3 & & \end{array}$	XeO ₆ ⁴⁻ + Xe(g) + O ₂ (g) + H ₂ O (two gaseous product)
50.	Hydrolysis of carbides. Be ₂ C Al ₄ C ₃ CaC ₂ Na ₂ C ₂ Mg ₂ C ₃	2Be(OH) ₂ + CH ₄ 4Al(OH) ₃ + 3CH ₄ Ca(OH) ₂ + C ₂ H ₂ NaOH + C ₂ H ₂ Mg(OH) ₂ + C ₃ H ₄
51.	Hydrolysis of phosphides M ₃ P ₂	3M(OH) ₂ + 2PH ₃
52.	Hydrolysis of nitides M ₃ P ₂	3M(OH) ₂ + 2PH ₃
53.	Hydrolysis of sulphides	

	M_2S_3 (M = Cr, Al)	$2M(OH)_2 + H_2S$
54.	CaNCN	$CaCO_3 + NH_3$ (Through urea formation $NH_2 - \overset{O}{\parallel} C - NH_2$)
55.	$B_3N_3H_6$	$B(OH)_3 + 3NH_3 + H_2$
56.	SiH_4	$SiO_2 \cdot nH_2O + 4H_2$



PW Web/App - <https://smart.link/7wwosivoicgd4>
 Library- <https://smart.link/sdfez8ejd80if>

DPP-13_Chemical Bonding_Chemistry(IOC)_10th Class

Draw the structure and predict the hybridisation of central atom and shape of the species.

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| <p>1. SnCl_3^- (trichlorostannite)</p> <p>2. HPO_3^{2-} (phosphite) ion</p> <p>3. SO_3^{2-} (sulphite)</p> <p>4. O_3 (ozone)</p> <p>5. $\text{Cl}_3\text{C}_3\text{N}_3$ (cyanuric trichloride)</p> <p>6. $\text{C}_3\text{N}_3(\text{N}_3)_3$ (cyanuric triazide)</p> <p>7. $\text{C}_3\text{N}_3(\text{NH}_2)_3$ (cyanuric triamine) (melamine)</p> <p>8. HSO_3Cl (Chlorosulphonic acid)</p> <p>9. $\text{I}_3^-, \text{I}_5^-$</p> <p>10. $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (blue vitriol)</p> <p>11. $\text{Na}_6\text{P}_6\text{O}_{18}$ (sodium hexametaphosphate)</p> <p>12. $\text{H}_5\text{P}_3\text{O}_{10}$ (tripolyphosphoric acid)</p> <p>13. $\text{H}_6\text{P}_4\text{O}_{13}$ (tetrapoly phosphoric acid)</p> <p>14. $(\text{CN})_2$ and $(\text{SCN})_2$ (cyanogen gas and thiocyanogen gas)</p> <p>15. P_4S_{10}</p> <p>16. Ba_2XeO_6 (barium perxenate)</p> <p>17. $\text{Ba}(\text{MnO}_4)_2$</p> <p>18. CsH_2PO_2 (cesium hypophosphite)</p> | <p>19. NaIO_3 (sodium iodate)</p> <p>20. SCN^- (thiocyanate ion)</p> <p>21. NH_4OCN (ammonium cyanate)</p> <p>22. $\text{Na}_2\text{S}_4\text{O}_6$ (sodium tetrathionate)</p> <p>23. $\text{H}_2\text{P}_2\text{O}_5^{2-}$ (pyrophosphite)</p> <p>24. $\text{H}_2\text{S}_2\text{O}_5$ (pyrosulphorous acid)</p> <p>25. $\text{H}_2\text{S}_2\text{O}_3$ (thiosulphuric acid)</p> <p>26. H_2PO_3^- (hydrogen phosphite ion)</p> <p>27. $\text{H}_2\text{P}_2\text{O}_7^{2-}$ (dihydrogen pyrophosphate)</p> <p>28. $\text{Na}[\text{B}_3\text{O}_3(\text{OH})_4]$</p> <p>29. $\text{Na}_2[\text{B}_2\text{O}_4(\text{OH})_4]$ (sodiumperoxyborate)</p> <p>30. $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$ (borax)</p> <p>31. Inorganic graphite $(\text{BN})_x$</p> <p>32. CaCN_2 (calcium cyanamide)</p> <p>33. Cl_2O (dichlorine monoxide)</p> <p>34. $\text{B}_3\text{N}_3\text{H}_6$ (inorganic benzene)</p> <p>35. CaC_2 (calcium carbide)</p> <p>36. $\text{H}_4\text{P}_2\text{O}_8$ (peroxidiphosphoric acid)</p> <p>37. $\text{H}_2\text{N}_2\text{O}_2$ (hyponitrous acid)</p> <p>38. $\text{H}_2\text{S}_2\text{O}_7$ (oleum)</p> |
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| 39. H_2SO_5 (peroxomonosulphuric acid) (Caro's acid) | 49. K_3IO_5 | 50. H_5IO_6 |
| 40. $\text{H}_4\text{P}_2\text{O}_6$ (hypophosphoric acid) (tetrabasic) | 51. N_2F_4 | 52. N_2F_2 |
| 41. Calcium phosphite (CaHPO_3) | 53. Hyposulphuric acid ($\text{H}_2\text{S}_2\text{O}_6$)/dithionic acid | |
| 42. Carborundum and Borazon [SiC and $(\text{BN})_x$] | 54. Isohypophosphoric acid (it is tri basic) | |
| 43. Dithionate and Dithionite ion ($\text{S}_2\text{O}_6^{2-}$ and $\text{S}_2\text{O}_4^{2-}$) | 55. Marshall's acid ($\text{H}_2\text{S}_2\text{O}_8$) | |
| 44. Ammonium phosphite [$(\text{NH}_4)_2 \text{HPO}_3$] | 56. SNF_3 | 57. XeF_3^- |
| 45. Sodium dihydrogen phosphate (NaH_2PO_4) | 58. P_4O_6 | 59. P_4O_7 |
| 46. Sodium dihydrogen pyrophosphate | 60. P_4O_8 | 61. P_4O_9 |
| 47. XeF_3^+ | 48. XeOF_4 , $[\text{XeF}_5]^+$ | 62. P_4O_{10} |

DPP-13_Chemical Bonding_Chemistry(IOC)_10th Class

ANSWER KEY

